# CSE 4511: Computer Networks

**QUIZ 1** **WINTER SEMESTER, 2024-2025**
**DURATION: 25 MINUTES** **FULL MARKS: 20**

## STUDENT ID:

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1. Suppose users share a 5 Mbps link. Each user requires 1 Mbps when transmitting but is idle 40 percent of the time. For the link, if the congestion is below 20%, it is considered working good enough. Given the scenario, when circuit switching is used, how many users can be supported? Also, when packet switching is used, how many users can be supported? Show full calculations. Assume that each users active/idle periods are independent of the others. $1 + 5 + 2$ (CO1) (PO1)

   Given,
   $X \equiv$ Number of devices that are active at the same time
   $P(X = k) = {}^nC_k(p)^k(1-p)^{n-k}$

**Solution:**

**Circuit Switching**
Since the total link capacity is 5 Mbps, the maximum number of users $N_{\text{circuit}}$ that can be supported is

$$N_{\text{circuit}} = \left\lfloor \frac{5\text{ Mbps}}{1\text{ Mbps/user}} \right\rfloor = 5.$$

Hence, *with circuit switching*, we can support exactly $\boxed{5}$ users.

**Packet Switching**
In packet switching, each user sends packets when active at 1 Mbps, but is idle 40% of the time (active 60% of the time). We assume users become active independently.

**Congestion Probability**
The probability of congestion is

$$P(\text{congestion}) = P\big(X > 5\big) = 1 - P\big(X \le 5\big) = 1 - \sum_{k=0}^{5} \binom{N}{k} (0.6)^k (0.4)^{N-k}.$$

We require

$$P\big(X > 5\big) < 0.20.$$

Equivalently,

$$\sum_{k=0}^{5} \binom{N}{k} (0.6)^k (0.4)^{N-k} > 0.80.$$

**Numerical Evaluation**
We check successive integer values of $N$ until the congestion probability exceeds 20%. Below is a small table showing $P(X > 5)$ for $N = 5, 6, 7, 8$. (Can compute these binomial sums with a calculator)
Interpretation:

- $N = 5$: $P(X > 5) = 0$.
- $N = 6$: $P(X > 5) \approx 0.0467 < 0.20$.
- $N = 7$: $P(X > 5) \approx 0.1586 < 0.20$.
- $N = 8$: $P(X > 5) \approx 0.3154 > 0.20$.

Thus, the *largest* integer $N$ satisfying $P(X > 5) < 0.20$ is

$$N = 7.$$

**Rubric:**

- 1 point for correct circuit switching answer
- 5 points for the correct congestion probability equation
- 2 points for correct numerical answer

2. CSMA/CA is used in IEEE 802.11 as the legacy MAC protocol. $3 \times 2$ (CO1) (PO1)

   a) Why CSMA/CA is used? Why not other MAC protocols?

   **Solution:**

   Wireless signal fades way with distance, so impossible to say of detecting collusion.

   b) How does RTS-CTS exchange ensure collision avoidance?

   **Solution:**

   RTS-CTS reserves the channel by alerting all nodes within range of both the sender and receiver to remain silent for the duration of the transmission, preventing hidden nodes from causing collisions.

3. Consider a router that interconnects three subnets: Subnet 1, Subnet 2, and Subnet 3. Suppose all of the interfaces in each of these three subnets are required to have the prefix 223.1.17 with subnet mask 24. Also suppose that Subnet 1 is required to support at least 60 interfaces, Subnet 2 is to support at least 90 interfaces, and Subnet 3 is to support at least 12 interfaces. Provide three network addresses (of the form a.b.c.d/x) that satisfy these constraints. $2 \times 3$ (CO1) (PO2)

**Solution:**

subnet 1 - 223.1.17.128/26
subnet 2 - 223.1.17.0/25
subnet 3 - 223.1.17.192/28